Crop Yield Prediction for Food Security in Africa

Improving Food Security in Africa Through Data-Driven Farming

Presented by
Epochs

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INTRODUCTION

Agriculture is vital for food production and the economy in many African countries. However, farmers often struggle to predict crop yields before planting, leading to poor planning, lower productivity, and financial losses.

The aim of this project was to develop a machine learning model that predicts crop yield using soil properties (pH, nitrogen, phosphorus, potassium, soil moisture) and field size, helping farmers make informed decisions and improve food security.

Success metrics:

- Lowest Mean Absolute Error (MAE).
- Lowest Root Mean Squared Error (RMSE).
- Highest R2 score.

STAKEHOLDERS

The system is designed to benefit everyone involved in agriculture, including farmers, government institutions, NGOs, research institutions, food security agencies, as well as consumers.

BUSINESS PROBLEM

- Globally, 2.33 billion people experienced moderate or severe food insecurity in 2023
- Africa has the world's highest undernourishment rate, affecting about 20% of its population
- Agricultural productivity remains low due to climate change, limited technology adoption and inadequate farming information
- In Kenya, about 98% of crop production depends on rainfall, making farming highly vulnerable to climate variability
- Many farmers still rely on experience rather than data when selecting crops, leading to lower yields.
- Farmers often plant without knowing their likely harvest leading to poor planning, thus leaving farmers exposed to financial loss and governments unable to plan for food needs.
- There is a need thus for decision-support tools to improve crop selection, productivity, and food security

PROJECT OBJECTIVES

Main Objective:

To support farmers in selecting suitable crops and improving agricultural productivity through informed decision-making.

Specific Objectives:

- Estimate the expected harvest of different crops .
- Identify crops that are best suited to different farming conditions.
- Provide farmers with alternative crop options to improve production.
- Support farmers in making informed crop selection decisions.
- Promote higher agricultural productivity and sustainable farming practices

DATASET OVERVIEW & DATA PREPARATION

13,802
Total Rows

11,783
Cleaned Rows

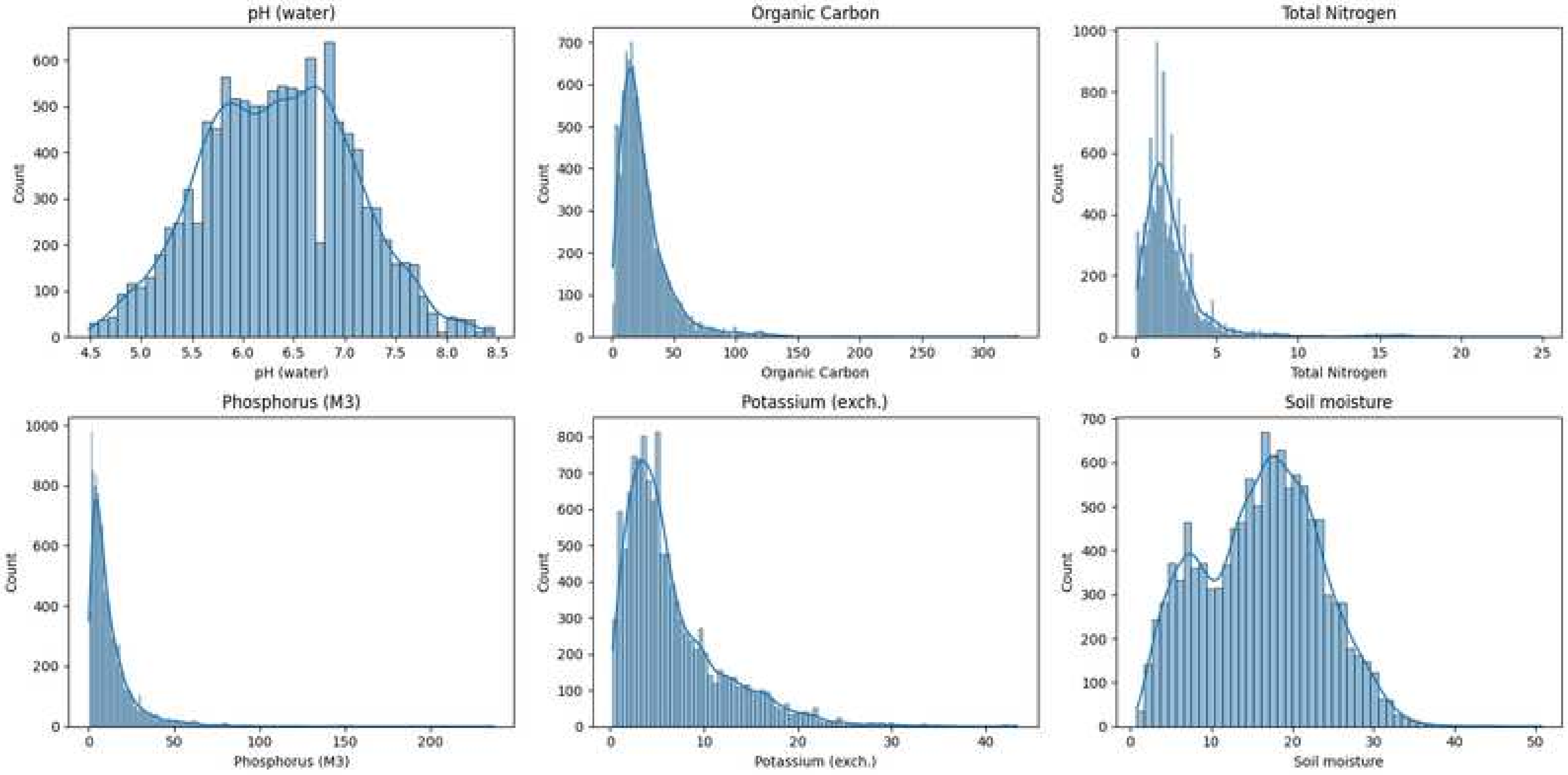
82
Crop Types

11
Total Columns

The dataset was obtained from an AgriTech company called Ycenter Shambah Solutions. Data cleaning entailed removal of duplicate rows, filling of missing values and handling of outliers.

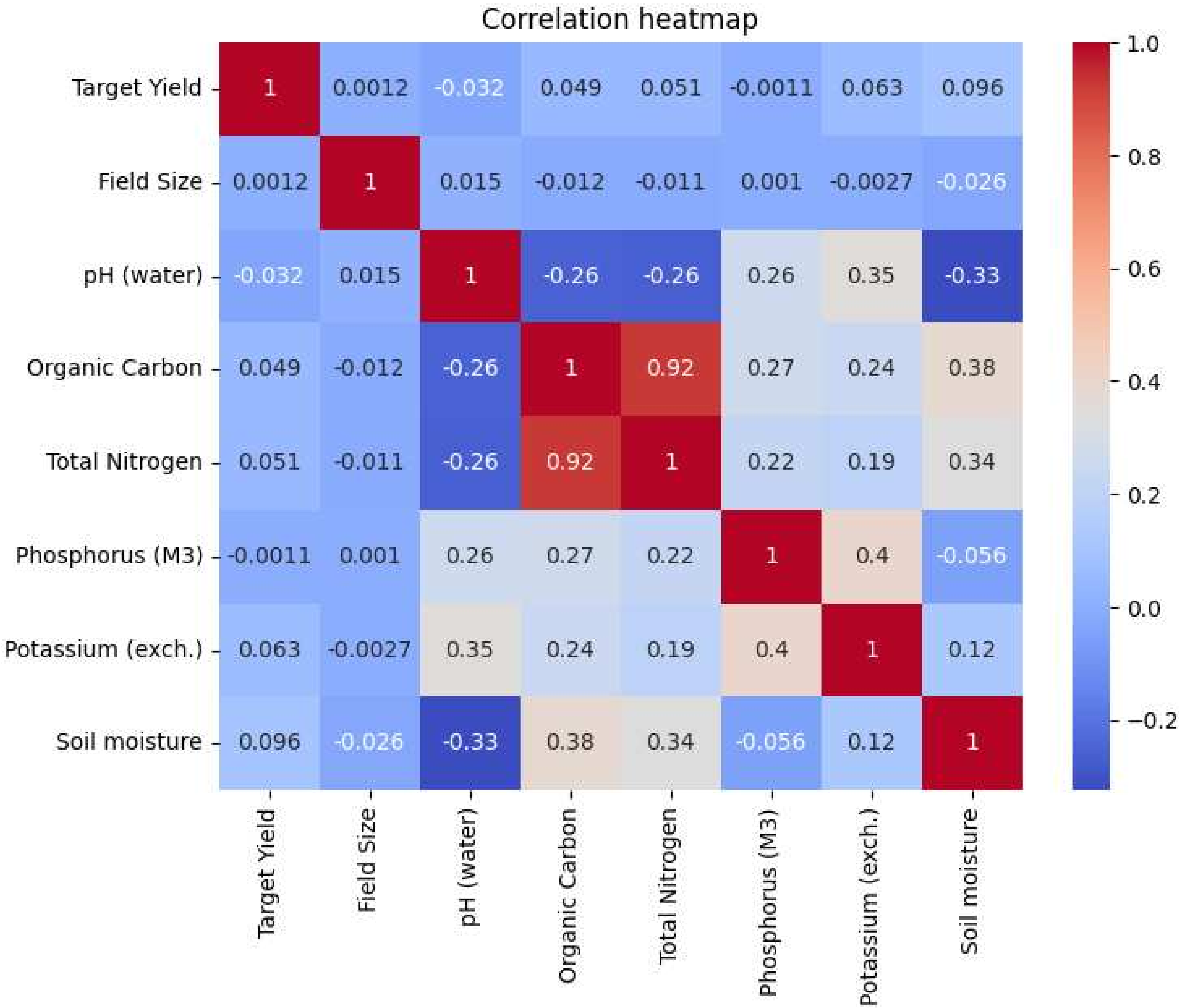
Crop Yield was used as the target variable, while soil properties, field size, and crop type were used as predictors. Before training the models, the target variable was log-transformed, crop names were one-hot encoded, and the data was split into training and testing sets, with feature scaling applied to the numerical variables.

EDA 1 – UNIVARIATE



The pH (water) variable exhibited an approximately normal distribution, while Organic Carbon, Total Nitrogen, Phosphorus (M3), and Potassium (exchangeable) showed pronounced positive skewness with long right tails.

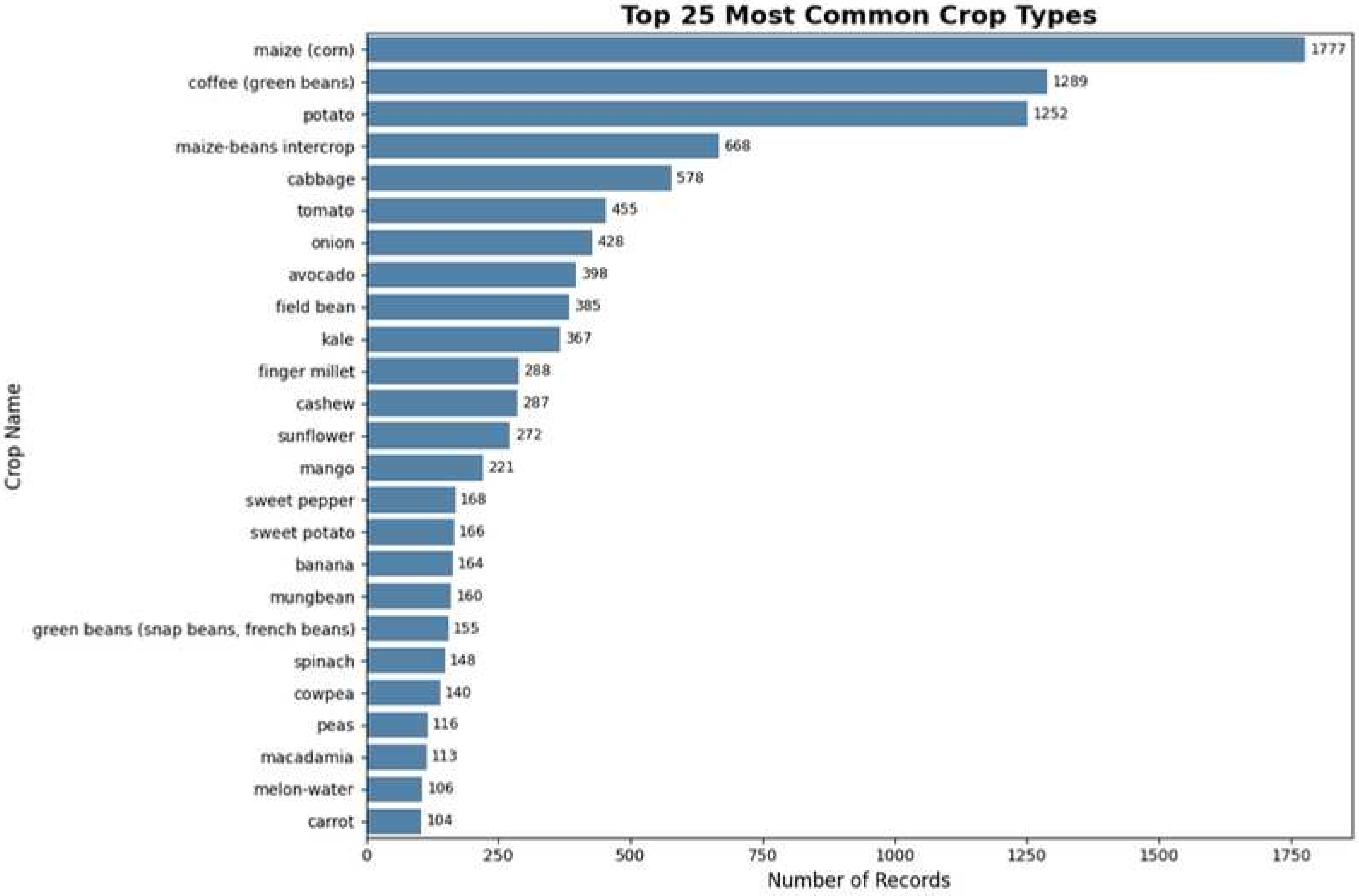
EDA 2 - CORRELATION HEATMAP



Organic Carbon and Total Nitrogen have a strong positive correlation of 0.92.

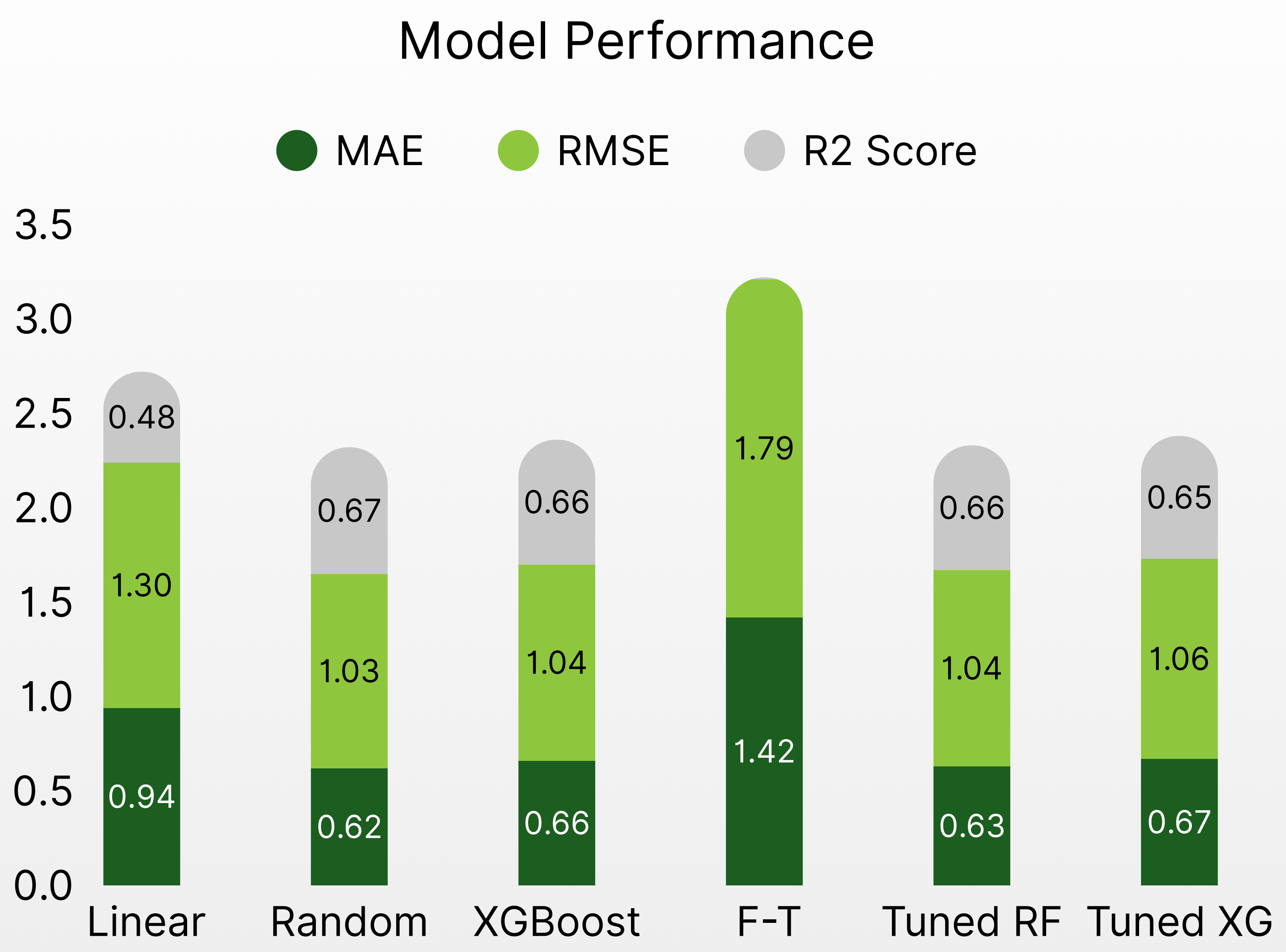
The Target Yield variable has a weak linear correlations with the numerical soil features. This suggests that crop yield is influenced by a combination of factors rather than a single soil property.

EDA 3 - MOST COMMON CROP TYPES



Maize had the highest number of records followed by green beans and potato with the least being melon-water and carrot respectively within top 25.

MACHINE LEARNING MODELS



Six models were evaluated to determine the most accurate approach for predicting crop yield. Linear Regression was used as the baseline model.

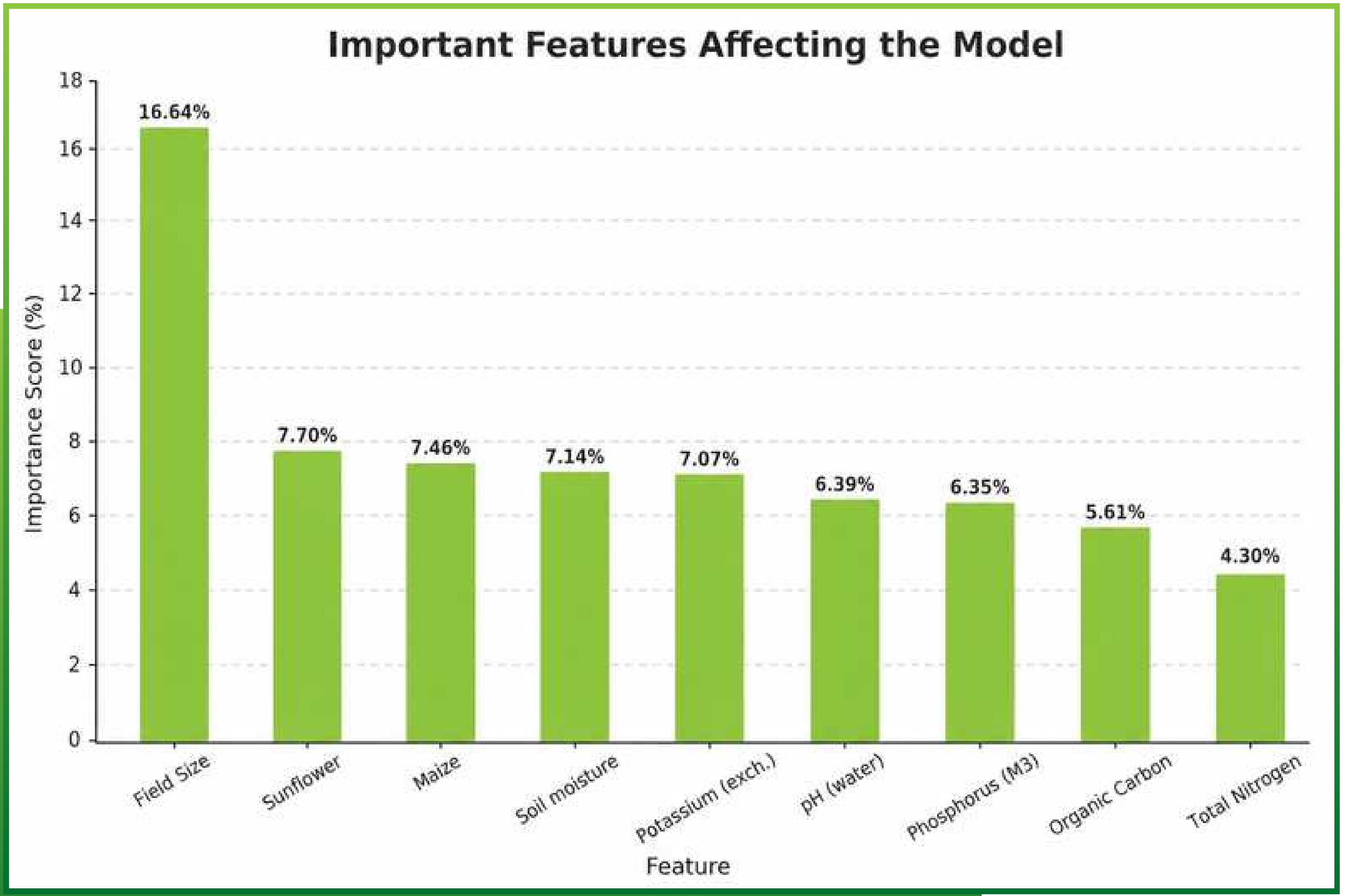
The models were compared using MAE, RMSE, and R^2 Score.

Random Forest achieved the lowest prediction errors and the highest R^2 score, making it the best-performing model for this project.

SUCCESS METRICS

Model	MAE	RMSE	R2 score
Linear Regression	0.94	1.3	0.48
Random Forest	0.62	1.03	0.67
XGBoost	0.66	1.04	0.67
F-T Transformer	1.41	1.78	0.02
Tuned Random Forest	0.63	1.04	0.66
Tuned XGBoost	0.67	1.05	0.66

MODEL INTERPRETATION



Feature importance was analyzed from the Random Forest model.

Field size had the greatest influence on crop yield predictions Other key factors included crop type, soil moisture, soil nutrients, and soil pH.

Crop yield is determined by a combination of factors, not a single variable

CROP RECOMMENDATION SYSTEM

Farmers first enter their soil test results, field size, and the crop they intend to plant. The system predicts the expected yield for that crop.

SELECT CROP

ENTER SOIL DATA

It then compares the same soil conditions across all other crops in the dataset and recommends a crop that is predicted to produce a higher yield.

PREDICT YIELD

GET RECOMMENDATION

SYSTEM DEMONSTRATION

Click the button below to see the system live

Live Demo

FINDINGS

- The Soil Moisture feature contained approximately 30% missing values while other have less than 1% missing value
- Random Forest produced the best Soil Moisture imputation performance for the 30% missing value since it outperforming KNN and XGBoost.
- Median imputation was used for remaining missing values after evaluating different methods.
- Applying a log transformation to the target yield reduced skewness, resulting in a more normally distributed target variable that is easier for machine learning models to learn.
- Crop yields are not evenly distributed; most observations fall within lower yield ranges while a few extreme values exist.

RECOMMENDATIONS

- Farmers should use this prediction model before planting to estimate the expected crop yield based on their soil properties and field size in order to make informed decisions.
- Farmers should also use the recommendation system to compare different crop options and select crops that are predicted to produce higher yields under the same soil conditions.
- Agricultural extension officers can use this system to provide farmers with data-driven advice on crop selection instead of relying only on experience.
- Soil testing should be carried out before planting since soil properties such as soil moisture, pH, organic carbon, nitrogen, phosphorus, and potassium were found to have a significant influence on crop yield.
- Government agencies and agricultural organizations can integrate this model into digital farming platforms to support food security planning and improve crop production.

THANK YOU

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